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Deuterium permeation through the low-activated V–4Cr–4Ti alloy under plasma irradiation

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ABSTRACT

Vanadium-based alloys are considered as candidate structural materials for nuclear fusion and fast nuclear reactors. Some properties of vanadium alloys are especially advantageous for nuclear fusion facilities, namely, a fast decay of induced activity, resistance to liquid lithium media and high-temperature strength. At the same time, V-alloys absorb and dissolve hydrogen isotopes in large quantities, and the mobility of dissolved hydrogen is high even at cryogenic temperatures. Thus, vanadium and its alloys, as some other metals of the fifth group (Nb, Ta), are characterized by high hydrogen permeability. This allows their usage as membranes for the diffusive purification of hydrogen isotopes. Under plasma irradiation of an inlet surface of such membranes, suprathermal hydrogen atoms and ions with quite high possibility penetrate through the adsorption barrier and diffusing inside materials bulk can reach an outlet surface. This paper aims to study the plasma-driven permeation (PDP) of deuterium through V–4Cr–4Ti alloy. It was shown that the permeation probability of deuterium under plasma irradiation for V–4Cr–4Ti alloy varies in a range of $\chi = 0.03 \div 0.12$ for $T = 600 \div 800$ K, which leaves open the possibility of such alloys usage for hydrogen isotopes purification in the fusion reactor exhaust.

1. Introduction

Hydrogen isotopes interaction with vanadium alloys has been actively studied over a number of decades. Vanadium alloys as well as some other metals (Nb, Ta) are characterized by high coefficients of hydrogen diffusivity and solubility. High hydrogen permeability of vanadium alloys is rather a disadvantage in terms of their usage as structural materials for fusion reactors, which can be exposed to tritium. However, this feature is very attractive for a number of other applications, also for the fusion needs.

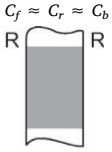
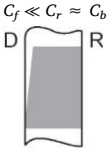
Essential interest to the alloys of fifth group metals (V, Nb, Ta) results from the perspectives of their usage as highly-selective membranes for diffusion purification of hydrogen isotopes from the gas phase [1–3]. However, due to a high hydrogen solubility of the fifth group metals a dangerous hydrogen concentration in them can be achieved at a relatively low hydrogen isotope pressures. One should mention that hydrogen embrittlement of materials is a rather serious problem for many branches of engineering and industry [4], but it is especially acute for alloys of the fifth group. For example, a ratio between hydrogen and metal atoms above ≈0.2 C(H/M) according to [1,3,5] could cause, for several reasons [6], significant decrease in strength, ductile-to-brittle transition and even failure of the fifth group metal specimens. For example, for pure vanadium [1] such ratio could be reached at $T = 673$ K and at hydrogen pressure $\geq 4 \cdot 10^4$ Pa, while in lower temperature region this pressure reduces significantly and at 423 K such ratio can be reached at ≥20 Pa [7,8]. Addition of elements like Mo and Al [9], W [10], Pd [8], Fe and Co [11], some others, and their combinations [12], to V and Nb allows to reduce hydrogen solubility in them and to increase the operating pressure values. Finally it leads to augmentation of hydrogen purification system performance and reliability [1,11]. Many systematic studies of hydrogen-selective alloy membranes are carried out and focused on finding the optimal composition of alloys to maintain high permeability and to prevent hydrogen embrittlement of them.

Series of studies in this field are devoted to investigation of hydrogen isotopes permeation through the fifth-group metals (V, Nb, Ta) under suprathermal hydrogen exposition [13–16]. Such experiments could be conditionally subdivided to the: plasma driven permeation (PDP), ion driven permeation (IDP) and permeation under neutral atoms irradiation (ADP). Taking into account some assumptions,

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Table 1
Suprathermal hydrogen permeation modes [17].

Permeation mode	*RR*	*RD*	*DR*	*DD*
Concentration relations	$C_f \approx C_r \approx C_b$	$C_f \approx C_r \gg C_b$	$C_f \ll C_r \approx C_b$	$C_f \ll C_r \gg C_b$
Concentration distribution				
Relation between incident and permeating fluxes	$J_p = \frac{k_b}{k_b + k_f}\alpha \cdot J_i$	$J_p = \frac{D_b}{L\sqrt{k_f}}\sqrt{\alpha \cdot J_i}$	$J_p = \frac{r^2 \cdot k_b}{D_f^2}(\alpha \cdot J_i)^2$	$J_p = \frac{D_b \cdot r}{D_f \cdot L}\alpha \cdot J_i$

J_p – permeating flux, J_i – incident particles flux, α – absorption/implantation coefficient, D – diffusivity, k – recombination coefficient, r – implantation depth, L – specimen thickness.

suprathermal hydrogen permeation for several extreme permeation modes summarized in the Table 1 could be described analytically [17–19]. Letters *R* and *D* means that permeation through the specimen could be limited by the recombination processes on surfaces (assuming fast diffusion), or by diffusion processes through the specimen (assuming fast recombination processes). Indexes *f* (face) and *b* (back), also as the first and second letters in permeation mode designation, are referred to the inlet and outlet surfaces respectively. One can see, that only in two modes permeating flux is a linearly dependent on incident particle flux – *RR* and *DD* modes.

For the fifth–group metals term "superpermeation", introduced for the first time by Livshits A.I. [20], is quite descriptive and means the ability of materials to transmit essential hydrogen fluxes comparable by magnitude with a flux of incident suprathermal atoms and ions. As it was mentioned above, the metals of the fifth group are characterized by quite big diffusivity and solubility of hydrogen, so a large number of hydrogen atoms can move quickly through the metal lattice. That is why RR should be the most ordinary permeation mode for such materials, and one can see that if $k_f = k_b$ (symmetric membrane case) $J_{p\ \max}$ could be up to 50% of the incident particles flux, while in case of asymmetric membrane $k_f \ll k_b$ permeating flux could be practically 100% of the incident flux (if $\alpha \sim 1$). In order to obtain such a high ratio of the permeating hydrogen flux to the incident one in the experiments, it's necessary to provide the following conditions: inlet surface should have a barrier transparent for suprathermal particles and reducing the inverse gas flux, while the outlet surface should not have such a barrier. Thin non-metallic films, (oxygen, carbon, sulfide and etc.), which can also be self-healing, can act as a barrier reducing the recombination on the inlet surface [15,21,22].

Taking into account the above, superpermeability of the fifth–group metals has a considerable importance for fusion applications due to perspectives of their usage for a high-selective pumping of hydrogen isotopes in demonstration and commercial fusion reactors [13,14,23] and as absorbing probes for edge plasma investigation [24,25]. Installation of niobium or vanadium membranes in the space between the torus vacuum chamber and pumping system (cryopumping) could allow to achieve a quite high specific pumping speed of $0.5 \div 2.5\ \mathrm{L{\cdot}cm^{-2}{\cdot}s^{-1}}$ in the pressure range of $10^{-3} \div 10^{-5}$ Pa [13,14] with an assembly of hot filaments (atomizers) usage. In this case the hydrogen compression reaches quite significant values from 10^3 [14] to $10^4 \div 10^5$ [13]. One can emphasize that specific pumping speeds for niobium and vanadium are comparable according to [14]. Also it's possible to install superpermeable membranes in the divertor region of the tokamak where low-density DT plasma and neutral atoms are present naturally. This approach is demonstrated in paper [26] with niobium membrane usage in JFT–2 M tokamak. The specific pumping speed for such system is about $\sim 0.45\ \mathrm{L{\cdot}cm^{-2}{\cdot}s^{-1}}$ at a pressure p ~ 0.03 Pa in the divertor region. It was also shown [26], permeating deuterium flux practically linearly depended on the pressure (and so on plasma density and, consequently, fluxes of incident particles), and increase with its rise. One should note that the selective pumping systems based on atomizers face the problem of the thermal sputtering and deposition of atomizer material on the inlet surface of superpermeable membrane [13]. In some cases it can adversely affect the symmetry of the superpermeable membrane and reduce system performance (e.g. W atomizer and Nb membrane). In this case one can speculate about using membranes and atomizers of the same material, which is possible for Nb and especially for Ta due to their high melting point. So, one can see that superpermeable membranes application in fusion reactors [13,14,26] could allow to reduce tritium fluxes pumped-out by cryopanels. This will increase safety and reduce tritium loads on the fuel cycle subsystems (tritium plant). It is especially important for the next generation fusion reactors such as DEMO, DEMO-FNS [27,28], due to significantly higher level of tritium consumption and circulation in them as compared with ITER.

One should also mention some features specific for niobium-based materials which can restrict their applicability in fusion reactors: activation under fast neutron irradiation, and about order of the magnitude higher hydrogen isotopes solubility at low temperatures compared with vanadium [7]. This in particular could lead to a hydride phase formation, a plasticity changing and even to the destruction of the superpermeable membranes in a case of a sudden sharp drop of the temperature. A neutron irradiation also causes negative changes in materials, for example, quite intensive swelling and embrittlement of pure vanadium [29,30]. Thus, it seems necessary for searching, developing and investigating acceptable superpermeable alloys which meet the requirements of fusion reactors according to several criteria: compatibility with fusion neutrons fluxes and fluences, constancy of thermal and mechanical properties under neutron irradiation. The V-based alloys like V–4Cr–4Ti are well-known structural materials candidates for fusion and fast neutron reactors. Fast decay of induced activity and durability in the liquid lithium media at high temperatures are the properties of vanadium alloys making them very attractive for fusion [31]. At the same time, vanadium alloys can absorb and dissolve hydrogen isotopes and the hydrogen flows through these materials can be high as compared with other structural materials.

Gas driven permeation (GDP) of hydrogen isotopes through vanadium and some V-based alloys is quite investigated [32–35], and was partially explored in our previous works [36]. At the same time, there are only several works with presented plasma-driven permeation data, some of which are discussed further, and the PDP data for V–4Cr–4Ti alloy is apparently absent. This paper presents some PDP and GDP data on deuterium with regard to V–4Cr–4Ti alloy, and its comparison with literature. Also possibilities of such alloy usage as a superpermeable membrane are discussed.

2. Experiment

The PDP of deuterium through V–4Cr–4Ti alloy was investigated with the help of PIM (magnetic plasma source) facility at the NRC «Kurchatov Institute» [37,38]. The scheme of the facility is shown in the Fig. 1. PIM facility consists of two main vacuum chambers which have separate evacuation systems – the exposition chamber *1*, where the

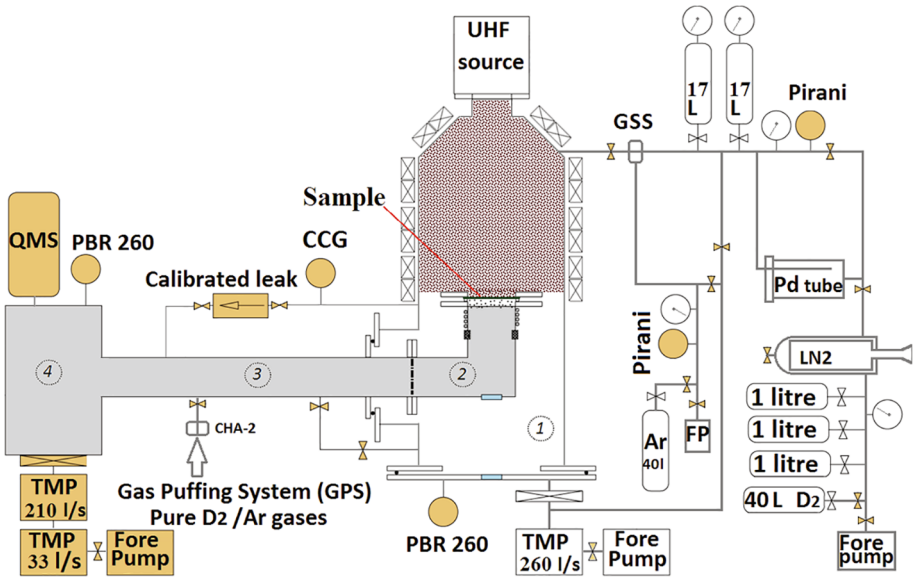


Fig. 1. The scheme of the PIM facility [37]: *1* – inlet (exposition) chamber; *2* – removable sample holder; *3* – intermediate volume; *4* – registration chamber; PBR 260 – full-range Pirani/Bayard-Alpert hot cathode vacuum gauge; CCG – ceramic capacitance vacuum gauge; QMS – quadrupole mass spectrometer; GSS – gas supply system.

plasma source is placed, and the registration chamber *4*. The sample holder *2* allows sealing the investigated sample which separates these two vacuum volumes.

The exposition chamber is a cylinder ~29 cm in diameter and ~60 cm in length. It is equipped with the ECR distributed plasma source which allows to obtain in a quite big volume a low-temperature plasma with temperatures of electrons and ions Te ~ 6 eV and Ti ~ 0.4 eV. One of the PIM facility discharge chamber peculiarity is a quite large area of homogeneous plasma (~2% deviation in density), with a diameter of about 150 mm. The ECR low-temperature hydrogen plasma usually consists of H_3^+, H_2^+ and H^+ ions, where the principal one is a H_3^+ molecular ion [39], and neutral atoms. Ion composition of PIM facility plasma source was previously investigated using a time of flight mass-spectrometer and carried out for protium (H_2). The H_3^+:H_2^+:H^+ ion relations are about 72.5 :21: 6.5% and correlate well with calculations and at the gas pressure $p = 6.1 \cdot 10^{-2}$ Pa and microwave power ~1 kW [37,38], and the relation of atomic and ionic hydrogen fluxes gave values about 30% in this regime according to calculations. A sample is electrically insulated from the holder and by BIAS potential application energies of incident ions can be varied in a range between ~4 eV/H (at floating potential) to hundreds eV/H. The density of the incident ion flux on the sample can reach 1 mA/cm^2.

The registration chamber is equipped with a quadrupole mass spectrometer (QMS) that registers partial pressures of some gases. For QMS ion current appropriate conversion into absolute values of the gas flux, calibration system is used, which allows to perform measurements with an error of 0.5–2% in a quite wide range of permeating fluxes [37,40].

Background pressures in registration and exposition chambers was about $1 \cdot 10^{-7}$ and $2 \cdot 10^{-6}$ Pa respectively. Deuterium, which was used in present experiments, was purified using the two-stage system consisted of the liquid nitrogen trap and a palladium tube filter for diffusion purification. In this paper gas and plasma driven permeation of deuterium with help of the differential method under continuous evacuation of the membrane outlet surface was investigated.

3. Specimen

This paper represents the data on deuterium permeation through V–4Cr–4Ti alloy, developed in Russian Federation [41].The Table 2 presents the chemical composition of V–4Cr–4Ti alloy, taking into account major impurities [42]. At the final production stage the material was underwent by manufacturer to the stabilizing annealing at $T = 1273$ K for 1 h in vacuum about 10^{-3} Pa. The specimens in the form of disks with a diameter of 60 mm were cut out from a hot-rolled foil with a thickness of 100 μm. The membrane *1* surface area which was exposed to a plasma was limited by a ring with inner diameter d = 47 mm as it shown in the Fig. 2. Specimen was sealed between holder *3* and pressing ring *2*, made of stainless steel, by argon arc-welding. One can see that pumping-out speed of the membrane outlet surface was limited by gas conductivity of the tube *4*, and can be estimated as ~3 L/s at room temperature. Some features related to sample sealing also with taking into account temperature distribution over the membrane radius were considered and described formerly elsewhere [36,37].

4. Investigation of deuterium permeation and discussion

Deuterium PDP of the V–4Cr–4Ti membrane was investigated after the GDP experiments, which were partially described elsewhere [36]. PDP experiments were carried out in the following manner:

- during continuous evacuation of the exposition chamber, inlet surface of the membrane is loaded by gaseous deuterium at a pressure of ~0.1 Pa;
- permeating gas flux reaches a steady-state value;
- initiation of the plasma discharge;
- applying BIAS potential to the specimen;
- a steady-state permeating gas flux obtaining at fixed conditions;
- series of the step by step temperature changing and corresponding steady-state permeating gas flux obtaining at each temperature;
- BIAS potential removal, plasma discharge disabling, gas supply shutoff. Specimen degassing and its subsequent cooling in vacuum.

At the Fig. 3 deuterium permeating fluxes after gas loading as a function of time, with subsequent plasma ignition and BIAS potential applying (–100 V) are presented.

After plasma discharge initiation deuterium fluxes increase and reach values exceeding by one or two order in magnitude ones obtained at permeation from the gas phase. After a BIAS potential applying (curve 2) with some delay local maximum and subsequent sharp decrease of permeating flux to steady-state value were observed. The latter is probably associated with a more intensive removal of the impurity (probably oxide) layer by energetic ions as compared with the irradiation at a floating potential (Ti ~ 4 eV/D) and a corresponding enhancement of recombination processes on a cleaner surface. One can

Table 2
Composition of V–4Cr–4Ti alloy (mass %) [42].

B	C	O	Ti	V	Cr	Fe	Si	Al	N	Cu	Ni + Nb + Mo	Co
0.002	0.013	0.02	4.36	Base	4.21	0.016	0.019	0.003	0.11	0.001	0.003	0.00003

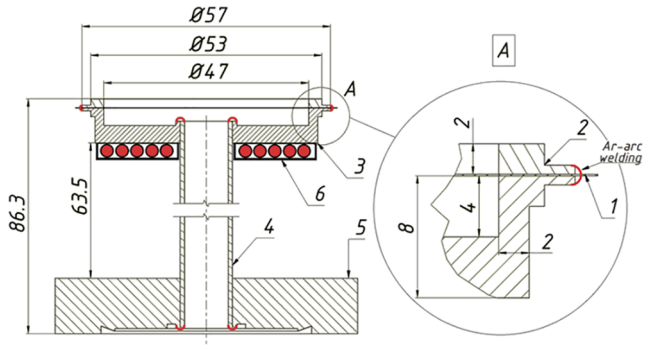


Fig. 2. Experimental specimen sealing scheme [37]: *1* – membrane; 2 – pressing ring. 3 – holder; 4 – tube Ø10 × Ø12 × 80 mm; 5 – DN40CF vacuum flange; 6 – insulated heater.

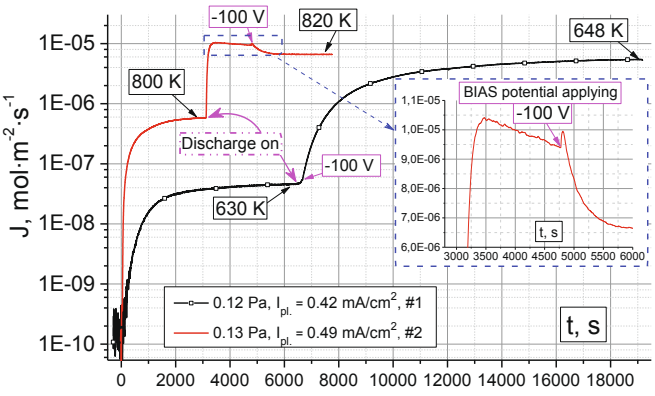


Fig. 3. Time dependencies of deuterium fluxes permeating through the V–4Ti–4Cr alloy.

see that in a case of the rapid BIAS potential applying after plasma discharge initiation (curve 1), such features are not observed on the permeation curve and resulting permeating flux of deuterium increases smoothly. Direct comparison of the curves also makes it possible to visually assess differences in activation energies of permeating fluxes dependence on temperature in case of GDP and PDP.

The Fig. 4 illustrates steady-state values of deuterium fluxes

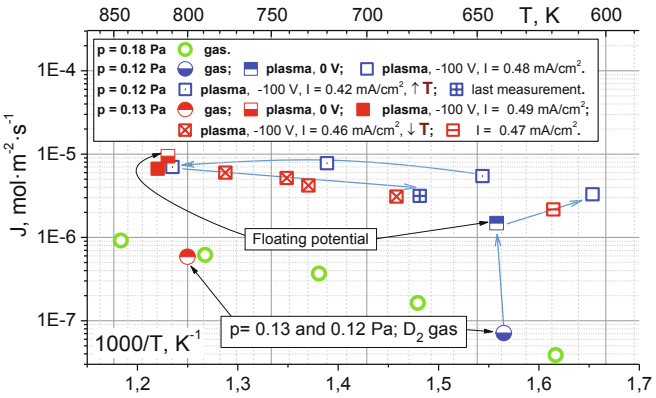


Fig. 4. Temperature dependencies of steady-state deuterium fluxes permeating through V–4Cr–4Ti under gas and plasma exposure.

permeating through the V–4Cr–4Ti membrane in case of gas (circles) and plasma (squares) exposure, as a function of a temperature. In this work, the data in the legends are ordered and depicted in the experiment conduction order.

Several experimental points were obtained at floating potential on the specimen, while the most of them were obtained under BIAS of –100 V. Such potential applied to the specimen allows to achieve an incident ion flux density I about ~0.5 mA/cm^2.

As it is shown in the Fig. 4, most PDP fluxes also have tended to growth with temperature increase, while the activation energy in this case is lower compared to GDP. One can see that obtained permeating flux values at the membrane temperature increase (↑ T) have a local maximum at a temperature of ~720 K, while in opposite case (↓ T) it's absent. Permeating fluxes in the case of ↑T at least 2 times higher as compared to the opposite case (↓ T), which cannot be explained as experiment error. The local maximum is more clearly depicted in the Fig. 5.

Such permeating flux behavior was also observed in hydrogen PDP studies carried out with niobium and was called hysteresis [38]. A nature of the observed local maximum can be explained by peculiarities of recombination processes on the membrane inlet surface. During an experimental PDP campaign, inlet surface was not sputtered by argon ions (cleaned), thus, some layer of impurities could be present on it. Also some impurities such as water vapor which are present in the residual gas of vacuum volumes, could affect the surface state and recombination processes on it. In this way, deuterium recombination and desorption from the inlet surface could be reduced, increasing thereby permeating flux. When T exceeds some value (here 720 K) the recombination processes seem to be accelerated essentially, and equilibrium impurity layer becomes thinner. This in turn leads to permeating fluxes reduction. It seems that for the series of measurements at the temperature decrease (red squares at the Fig. 4) the surface state and an oxide layer at 820 K remains unchanged when the temperature is going down.

Permeation probability χ is a quite descriptive and useful quantity for the interpretation of PDP experimental results, which can be written as:

$$\chi = J_{permeatingatoms}/J_{incidentatoms} = \chi_0 \cdot e^{\frac{-E_\chi}{RT}} \quad (1)$$

where χ_0 – pre-exponential factor, and E_χ – activation energy of permeation probability. As it was mentioned above, along with deuterium ions (mainly D_3^+ with Ti ~ 4 eV/D at floating potential) neutral atoms (up to 30% of the incident flux according to calculations by OGRAS code [37]) are also presented in ECR plasma of PIM facility. So, the ion current onto the specimen registered in experiments (e.g. I = 0.5 mA/cm^2) can be recalculated to the incident flux of deuterium atoms as: $J_{\ incident\ atoms} = I/e{\cdot}(n + 0{,}3) = 9.24{\cdot}10^{19}$ *at. D/(s·m^2)*, where e denotes the elementary charge, and n = 2.66 is a coefficient taking into account the concentration and composition of ions in a PIM facility plasma [37].

The Fig. 5 shows the permeation probability values as function on the inverse temperature. Designations are similar to those used in Fig. 4. One can see that obtained values of permeation probability χ for the investigated V–4Cr–4Ti alloy are in the range of 0.03 ÷ 0.12 for the corresponding temperature range 600 ÷ 825 K, and a local maximum in the case of temperature increase is more pronounced.

Fig. 5 also presents χ values for V–15Cr–5Ti alloy calculated on the base of the data obtained in the study [16]. Paper [16] includes some data of V–15Cr–5Ti membrane (500 μm in thickness, outer diameter – 25 mm which was significantly larger than ion beam diameter)

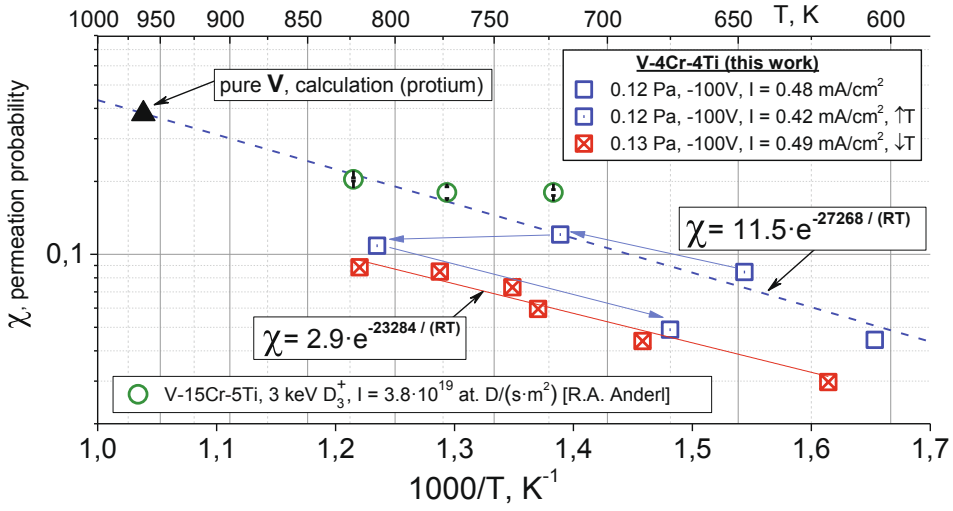
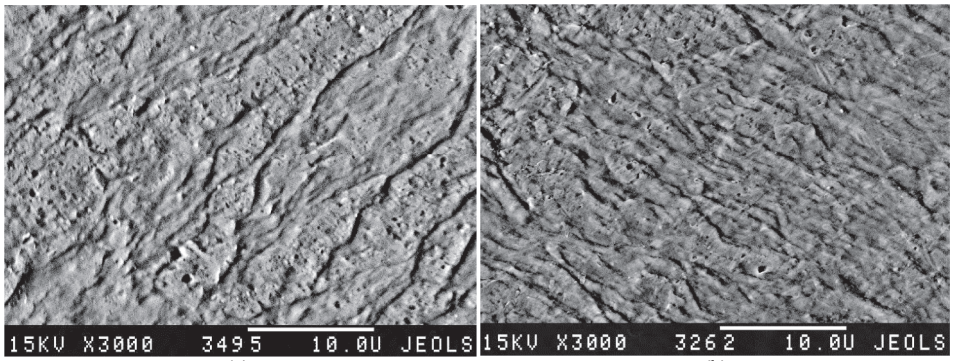


Fig. 5. Dependence of the deuterium permeation probability on the inverse temperature: for V–4Cr–4Ti membrane (this work), and for V–15Cr–5Ti alloy from [16], for comparison.

permeability under irradiation by D_3^+ ion beam with an energy of 3 keV and a flux density of $3.8{\cdot}10^{19}$ D at.$\cdot m^{-2}{\cdot}s^{-1}$. The authors of [16] concluded that the obtained permeability regime is closer to *RR* than to the *RD*. One can see that some data in this work (obtained at T increase up to 720 K) could be fitted like $\boldsymbol{11.5{\cdot}e^{-27268/(RT)}}$, and this fitting is correlate with a data from paper [16], despite sufficient differences in the organization of experiments. In addition, it would be quite curious to compare obtained experiment data with reliable literature for ADP and PDP of hydrogen for pure vanadium or its alloys. But unfortunately only several papers devoted to vanadium were found (most data is about niobium), in which data on hydrogen (protium) absorption coefficients ($\boldsymbol{\alpha}$) under ion [43] and atom [44] irradiation are presented. In both papers specimens with a thickness of 100 μm were examined. In the paper [43] PDP of oxygen-doped (2.9 at.%) vanadium was investigated at T = 963 K, while an incident H_2^+ ion energy was varied. A monotonous growth of absorption coefficients (from 0.24 up to practically 1) with energy increase (from 0.7 to 300 eV/H) was observed by authors. The authors assumed a symmetrical membrane in *RR* permeation regime, and equilibrium state between dissolved oxygen in the specimen bulk and it presence at the surfaces as a monolayer. In the case of atomic irradiation of vanadium [44] also assuming an equilibrium monolayer of oxygen on the surface, absorption coefficient was declared to be independent of the incident atomic flux density (in a range of $10^{17} \div 10^{20}$ H$\cdot m^{-2}{\cdot}s^{-1}$), and a monotonous growth of $\boldsymbol{\alpha}$ (0.15 ÷ 0.22) with temperature in a range of 343 ÷ 623 K was observed, while in T range 523 ÷ 623 K it seems to be saturated. On the basis of these papers, it's quite easy to estimate hydrogen permeation probability for vanadium in the case of PIM facility plasma. At such conditions $\boldsymbol{\chi = 0.38}$ at $\boldsymbol{T = 963}$ **K** and BIAS = −100 V, and this value is in a perfect agreement with a blue dash fitting presented in the Fig. 5.

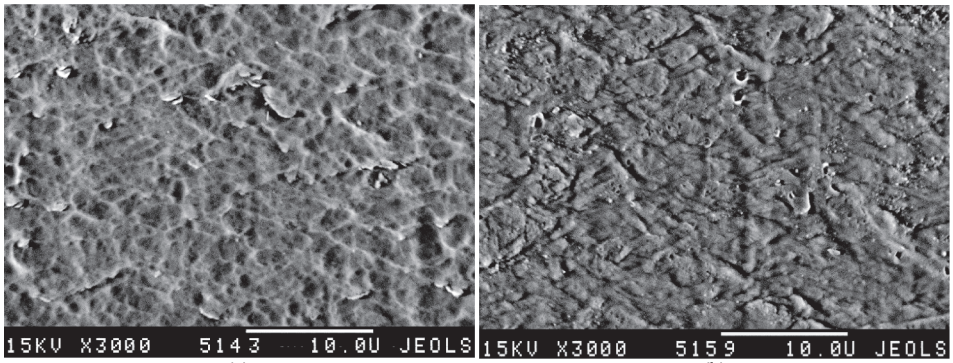
Due to the large number of papers devoted to niobium, it's quite curious to make such estimations also for Nb. According to [45] absorption probability of the hydrogen atoms for niobium is a constant value, about 0.21 (0.25 in paper [44]) and doesn't depend neither on oxygen concentration in the bulk, nor on temperature. While at BIASed plasma [15,46] relation of J_{BIAS}/J_{float} for Nb is decreased by about one order of the magnitude with BIAS increase (from floating potential to −250 V), except some cases: high temperatures and/or high oxygen concentrations in the bulk. As it can be seen from [15] in some cases (depending on temperature, and at not a maximum values of the BIAS) relation of J_{BIAS}/J_{float} could reach values of 1.2 ± 0.1, while according to [46], this relation is 1. Thus, taking into account published data, one can estimate from above permeation probability $\boldsymbol{\chi}$ as **0.11–0.16** in a wide temperature range, for PIM-like plasma, in the case of oxygen-doped symmetric niobium membrane, resistant to ion sputtering.

Lets return to discussion of results of the present work and to alternative explanation of the observed local maximum. It is well known that oxygen concentration in V-Cr-Ti alloys significantly depends on the heat treatment, and usually about 0.01 ÷ 0.18 at% [47–50] which is quite small as compared with pure vanadium, doped by oxygen, discussed above. In present work it is 0.06 at. %, see Table 2. Taking from the literature the assumption about equilibrium monolayer of the oxygen on the surface one can also suppose that under plasma irradiation oxygen can be sputtered from the surface or desorbed from the surface by recombination with hydrogen. As the result, one can expect impoverishment by oxygen of the near-surface region during irradiation and, subsequent acceleration of recombination processes on the clean surface with temperature increase. At the same time, for V-4Ti alloy it was shown [50] that during annealing in vacuum at T > 600 K relation of V/VO_x on the surface significantly increased with the annealing temperature rising, and this was explained by authors by oxygen diffusion into the bulk. At the same time [50] relation of surface concentrations Ti/TiO_x was a constant value up to 983 K, while at higher temperatures segregation of Ti occurred, and Ti concertation on the surface increased from 20 to 40 at.% with a corresponding T increase from 983 to 1223K. The segregated titanium could be easily oxidized by oxygen from impurities in the residual gas. So in temperature conditions of our work TiO_x will be present on a surface. It is well known that oxides significantly reduce fluxes permeating from gas phase and increase the permeability activation energy (E_p). For example in [51] authors attributed E_p increase from 14.9 to 46.6 kJ/mol and decrease of permeating flux by 2–4 orders of magnitude for the annealed sample of V-4Cr-4Ti alloy as compared with unannealed one, exactly by titanium segregation on the surface during vacuum annealing (T = 1000C, 2 h, p ~ 3e–5 Pa) and its subsequent oxidation by residual gases (H_2O, CO, etc.). So a nature of the local maximum of the permeation probability observed in our work can be also explained by reduction of an oxide layer due to deoxidation of vanadium at annealing (oxygen diffusion into the bulk), while titanium oxides remain stable. And a red fitting depicted in the Fig. 5 could be attributed mostly to titanium oxides on the surface. Probably titanium oxide is self-healing because of presence of oxygen in residual gases. Thus, it seems, that rather pronounced temperature dependence of permeation probability $\boldsymbol{\chi}$ (E_χ = 23 and 27 kJ/mol) could characterize outlet surface state (probably titanium and/or vanadium oxide layers and/or processes on it). Highly likely that in IDP experiments of V-15Cr-5Ti in [10], where area of ion irradiation was considerably smaller than area of membrane, oxide layer on a bigger part of the inlet surface remains stable and reducing the



(a) (b)

Fig. 6. Micrographs of the inlet – (a) and outlet – (b) surfaces of V–4Cr–4Ti specimen before the experiments.



(a) (b)

Fig. 7. Micrographs of the inlet – (a) and outlet – (b) surfaces of the investigated V–4Cr–4Ti specimen after the experiments on gas and plasma driven permeability.

backward desorbing flux. At the same time, a more significant addition of chromium can also lead to some changes in the state of the surface and surface processes, as well as changes in a bulk with respect to oxygen (e.g. χ value at 723 K in the Fig. 5).

5. Morphology of the specimen

The micrographs of the inlet and outlet surfaces of the V–4Cr–4Ti membrane before and after experiments (GDP and PDP) are presented at the Figs. 6 and 7, respectively.

As it was mentioned above [36] and can be seen from the Figs. 6 and 7, the relief of the inlet surface after irradiation by argon and deuterium ions was changed significantly, while the outlet surface of the membrane which was in ultrahigh vacuum conditions, remained almost unchanged. One should mention that inlet surface irradiation by Ar ions was used only during GDP sessions, which preceded PDP ones.

6. Conclusion

Based on the results obtained in this study it can be concluded that deuterium permeation probability χ of V–4Cr–4Ti alloy under BIASed ECR plasma irradiation is ranged like 0.03 ÷ 0.12 in the temperature range of 600 ÷ 800 K which is close to niobium but about 3 times smaller than for pure vanadium. Also, one can note rather pronounced temperature dependence of permeation probability χ with E_χ = 23 and 27 kJ/mol, which could characterize the outlet surface state (probably, titanium oxide layer presence) and/or processes on it. So, it can be concluded, that this alloy as some other V-based alloys can be considered as a super permeable membrane for the selective extraction of hydrogen isotopes for fusion devices. It is also important to emphasize that V–4Cr–4Ti alloy is more stable under neutron irradiation as compared with pure vanadium (it doesn't swell under neutron irradiation) and by contrast with niobium is a low-activated material with a fast decay of induced activity. Nevertheless systematic investigations devoted to the surface state and processes on it, especially taking into account fusion device exhaust composition, should be carefully performed.

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7. Data availability

Data will be available upon request.

CRediT authorship contribution statement

D.I. Cherkez: Writing - original draft, Formal analysis, Investigation, Conceptualization, Writing - review & editing, Visualization. **A.V. Spitsyn:** Funding acquisition, Investigation, Project administration, Conceptualization, Supervision. **A.V. Golubeva:** Validation, Writing - original draft. **V.M. Chernov:** Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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